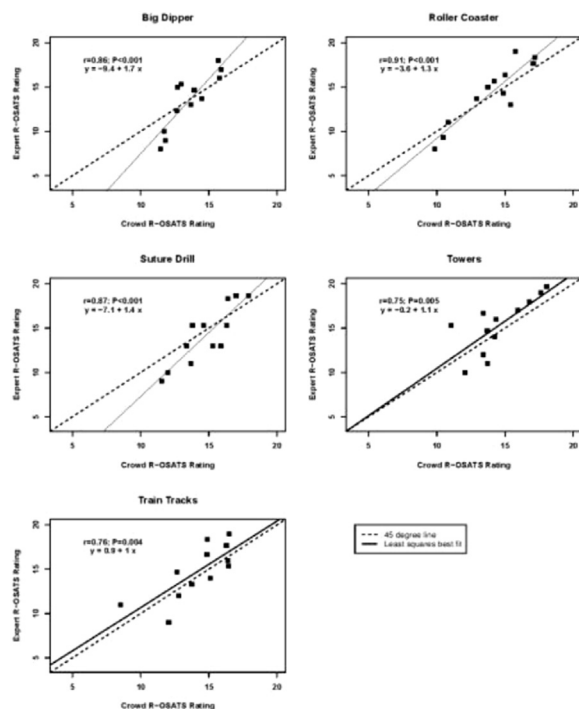




Conclusion: Crowdsourced assessments of recorded dry lab surgical tasks using a validated assessment tool are a suitable alternative to faculty expert evaluation. Crowdsourcing would provide a rapid and accurate method of assessing technical skills of trainees while minimizing burdens on faculty time.

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Laparoscopic and Robotic Skills Are Transferable in a Simulation Setting

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Study Objective: To assess the transferability of skills from robotic to laparoscopic simulators and vice versa among simulation naive participants.

Design: Randomized single-blinded controlled trial.

Setting: Johns Hopkins Bayview Surgical Simulation Center.

Patients: Forty medical students with limited prior laparoscopic and robotic simulation experience.

Intervention: Participants completed a baseline practice on a robotic pegboard transfer task (dV-Trainer, Mimic, Seattle, WA) and a laparoscopic peg transfer task (Fundamentals of Laparoscopic Surgery [FLS], VT Medical Inc., Waltham, MA). Skills were evaluated using validated objective and subjective measures (global rating scales (GRS)) by two blinded expert surgeons. Participants were then randomized to practice on either of the above robotic (N=20) or laparoscopic (N=20) task. After practice, participants were again evaluated performing both tasks.

Measurements and Main Results: At baseline, there were no significant differences in objective measures or composite GRS scores on both the robotic and laparoscopic tasks between the two groups. Participants in the laparoscopic practice group completed the final laparoscopic task faster (100 vs. 158 sec, $p<.003$) and with higher percent improvement in GRS scores (67.5% vs. 27%, $p<0.001$) compared to the robotic practice group. The robotic practice group performed the robotic task faster (71 vs. 120 sec, $p<0.001$) and more efficiently (economy of motion $p<0.001$), with higher percent improvement in GRS scores (77% vs. 29%, $p<0.001$) compared to the laparoscopic practice group. Participants who practiced laparoscopically did improve their robotic performance with an increase

in mean GRS scores of 15 to 20 but this was not statistically significant ($p=0.091$). Participants who practiced robotically did significantly improve their laparoscopic performance with an increase in mean GRS scores of 15 to 19 ($p=0.02$).

Conclusion: Medical students significantly improved their performances following practice on either robotic or laparoscopic simulators. Skills learned on either platform appears to be transferable; however, the amount of improvement was still significantly less than practice on the respective platforms.

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Development and Evaluation of a Low-Cost, Reusable Laparoscopic Entry and Emergency Model

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Study Objective: To develop and evaluate a low-cost laparoscopic emergency model.

Design: The study was designed to establish content, response process, and relations with other variables evidence using performance checklists during the scenario and post-simulation surveys.

Setting: Model development and testing was completed in a multidisciplinary simulation center at a tertiary care center.

Patients: A convenience sample of 29 gynecology, urology, and general surgery residents, fellows, and attending surgeons participated in the laparoscopic emergency scenario.

Intervention: A low-cost, laparoscopic entry model that contained a retroperitoneal hematoma hidden by simulated bowel was created.



Following laparoscopic entry, the simulated patient developed hemodynamic instability. A performance checklist was completed during the scenario to evaluate recognition and management of potential etiologies of hemodynamic instability following entry. Participants completed a post-simulation questionnaire on the realism of the model and scenario.

Measurements and Main Results: The bowel was retracted and the hematoma identified laparoscopically by 3 (25%) of 12 junior residents versus 4 (23.5%) of 17 senior residents, fellows, and attendings ($p=0.87$). A statistically significant difference ($p=0.005$) existed in the decision to perform an exploratory laparotomy in response to hemodynamic instability (41.7% in the junior resident group versus 82.4% in the senior resident, fellow, and attending group). Of the 29 participants, 27 (93.1%) agreed or strongly agreed that the simulated drill approximates the stress of a vascular injury during laparoscopy and 27 (93.1%) agreed or strongly agreed that the model set up appears appropriate for approximating a retroperitoneal hematoma.

Conclusion: The reusable, laparoscopic simulation model and emergency scenario were rated favorably by participants. The majority of participants did not retract the bowel and attempt to identify the simulated retroperitoneal hematoma laparoscopically. There was a statistically